

7.4 Deriving Area Formulas for Trapezoids, Parallelograms, and Rhombi

Lesson Introduction

 Benchmark	MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.		 Learning Target	I can use the formulas for the area of triangles and rectangles to derive formulas for the area of trapezoids, parallelograms, and rhombi.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		MA.912.GR.4.4. Solve mathematical and real-world problems involving the area of two-dimensional figures.	
 Essential Questions	How do you derive the area formulas for parallelograms, trapezoids, and rhombi by decomposing and rearranging their pieces?			
 Lesson Narrative	This is part one of a two-part lesson series on the areas of parallelograms, trapezoids, and rhombi. In this lesson, students derive the formulas for the area of parallelograms, trapezoids, and rhombi by decomposing and rearranging their pieces. This work will prepare students for the next lesson, where they will be applying the formulas to solve real-world problems.			
 Component Summary	7.4.1 Warm-Up (~5 min.)	Notice and Wonder about a decomposed rectangle (<i>SE p. 315</i>)	7.4.4 Activity (10 min.)	Decompose trapezoids to find the formula for area of trapezoids (<i>SE p. 318</i>)
	7.4.2 Refresher (5 min.)	Review decomposing to find area of a triangle (<i>SE p. 315</i>)	7.4.5 Activity (10 min.)	Decompose rhombi to find the formula for area of rhombi (<i>SE p. 319</i>)
	7.4.3 Activity (10 min.)	Decompose parallelograms to find the formula for area of parallelograms (<i>SE p. 316</i>)	7.4.6 Wrap-Up (~5 min.)	Self-reflection on what was most difficult or confusing (<i>SE p. 321</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> - Quadrilateral - Parallelogram - Trapezoid - Rhombus <u>Additional Terms:</u> N/A	(Optional) Graph paper (Optional) Colored highlighters (Optional) Scissors	Prepare copies of directions and problems for the Activities (7.4.3-7.4.5) and set up stations for each activity. Determine groups for rotations through the different stations. Consider using data and/or observations from formative assessments to determine grouping arrangements.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse which diagonal represents the base and which diagonal represents the height in the triangles created when decomposing a rhombus. - Students may not recognize that the bases in a trapezoid can be used interchangeably in the formula.	- The activities provide scaffolding, yet students may still struggle with deriving the formulas. Allow for some productive struggle. Students may struggle with the rhombus activity the most, so consider any extra help your students may need to complete the activity. - Students without a strong understanding of equivalent algebraic expressions may think their “formula” does not match their peer’s formula or any provided formulas in the unit. When supporting these students, use correct mathematical properties to explain why their formula is the same. For example, if a student has $\frac{1}{2}d_2d_1$ instead of $\frac{1}{2}d_1d_2$, explain that the commutative property shows the two are equivalent. Free students to use their own derived formulas for the rest of the unit. A student may understand $A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$ more than $A = \frac{1}{2}h(b_1 + b_2)$.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder	MA.K12.MTR.2.1: Multiple Representations
	The Warm-Up and Refresher purposefully position students to confidently share their observations about a decomposed rectangle. The discussion will lay the foundation for their explorations with decomposing parallelograms, trapezoids, and rhombi to derive their area formulas.	In the Activities (7.4.3-7.4.5), students decompose parallelograms, trapezoids, and rhombi. They rearrange the decomposed pieces to provide a concrete representation of the area formula for the figure. Through the activities, students are guided from concrete, to pictorial, to abstract representations as their understanding of area progresses.
Lesson Notes:		

7.4.1 Warm-Up: Notice and Wonder

Component Overview: Students reflect on what they notice and wonder about an image of a rectangle decomposed into two triangles and a trapezoid. Student thinking and reasoning in this Warm-Up lays the foundation for their work in the subsequent activities.



7.4.1 Warm-Up: Notice and Wonder

Two figures are shown.

Figure 1



Figure 2



1. What do you notice about these two figures?
Answers will vary. I notice that when combined, Figure 2 creates Figure 1.
2. What do you wonder about the figures?
Answers will vary. I wonder what shape is in the middle of the two triangles in Figure 2.



Notice and Wonder

There are no wrong answers. This fosters an environment where each student feels safe to participate. After sharing, students will likely feel more confident sharing their observations during their explorations as they derive the area formulas in subsequent activities.



How can you combine the two triangles in Figure 2 to make a smaller rectangle? How could the shape in the middle of Figure 2 be decomposed into two smaller triangles?

7.4.2 Refresher: Decomposing to Find the Area of a Triangle

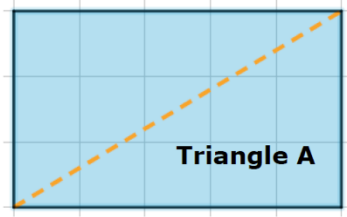
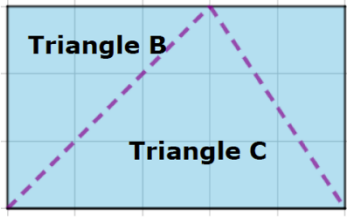
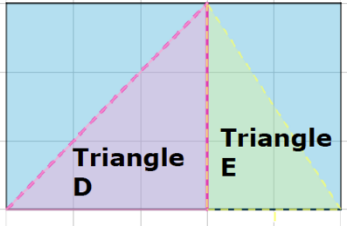
Component Overview: Students use three methods to decompose rectangles into triangles and then rearrange the triangles to find the area of the rectangle. Students can complete the table individually or with a partner. Debrief by engaging students with a whole-class discussion to ensure each and every student is ready for the subsequent activities.



7.4.2 Refresher: Decomposing to Find the Area of a Triangle

Figures can be decomposed into smaller pieces that are then rearranged and composed into other shapes. Decomposing is useful when finding the area of a figure.

1. Complete the table.

The rectangle is divided into 2 triangles. The area of Triangle A is $\frac{1}{2}$ of the area of the rectangle.	
The rectangle is divided into 3 triangles. The area of Triangle B is $\frac{3}{10}$ of the area of the rectangle. The area of Triangle C is $\frac{1}{2}$ of the area of the rectangle.	
The rectangle is divided into 4 triangles. The area of Triangle D is $\frac{3}{10}$ of the area of the rectangle. The area of Triangle E is $\frac{1}{5}$ of the area of the rectangle.	



Leverage this activity not as an opportunity to re-teach, but instead to highlight the utility of decomposition for finding area.



How do you know the area of triangle ____ is ____?
How could you check your answer?

7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram

Component Overview: The next three activities can be used as three different stations. Groups of students can rotate between the stations in any order. For large classes, two copies of each station can be used. In this first activity, students decompose a parallelogram into a trapezoid and a triangle. Then, they shift the triangle to form a rectangle and derive the area formula for a parallelogram.



7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram



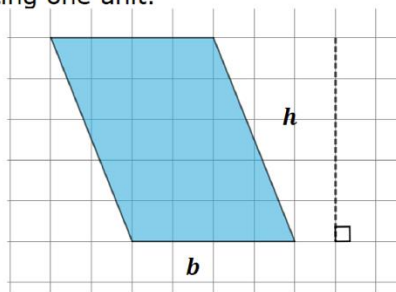
A **quadrilateral** is a polygon with exactly four sides and four vertices.

A parallelogram is a type of quadrilateral. Not only does a parallelogram have four sides and four vertices, but its opposite sides are also parallel.



A **parallelogram** is a quadrilateral containing two pairs of parallel sides.

1. A parallelogram is shown on a grid with each gridline representing one unit.



Complete each statement.

- a. The base of the parallelogram is 4 units.
- b. The height of the parallelogram is 5 units.



Consider your students and classrooms when planning the next three activities. Station rotations could be an interactive way for students to stay active while working.

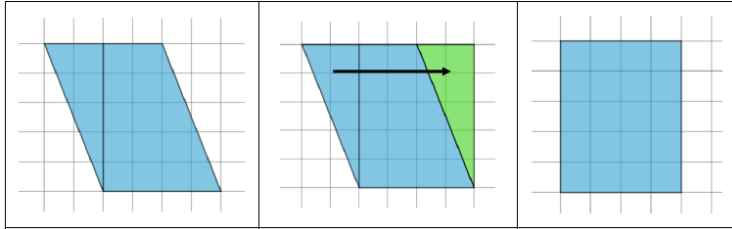
- Station 1: Activity 7.4.3 – Parallelogram
- Station 2: Activity 7.4.4 – Trapezoid
- Station 3: Activity 7.4.5 - Rhombus



Provide a kinesthetic entry point by having students physically draw, cut, and rearrange the parallelogram. Provide materials at each station or elect a “Materials Captain” per group to handle the supplies for each group.

7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram (continued)

2. The images below show the same parallelogram being decomposed and then rearranged to form a rectangle.



Complete each statement.

- The base of the rectangle is 4 units.
 - The height of the rectangle is 5 units.
3. Discuss with your partner the relationship between the area of the parallelogram and the area of the rectangle. Write a summary of your discussion.
Answers will vary. The relationship between the area of the parallelogram and the rectangle is that the areas are the same.
4. Complete the statements.

A parallelogram can be decomposed and rearranged into a

☐ triangle
☒ rectangle

with the same base and height. The

formula used to find the area of a rectangle is

☐ $A = bh.$
☒ $A = \frac{1}{2}bh.$

Therefore, the area of the parallelogram is

☐ $A = bh.$
☒ $A = \frac{1}{2}bh.$



Consider having students create a foldable and/or an Anchor Chart with images and examples of deriving each area formula to provide a visual entry point.

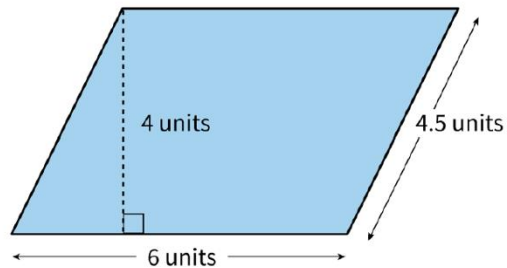


Consider asking students the following:

- How does the height of the parallelogram compare to the height of the rectangle?
- How does the base of the parallelogram compare to the base of the rectangle?
- How do these relationships help you to determine the area of the parallelogram?

7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram (continued)

5. A parallelogram with its dimensions labeled is shown.



Students may use 4.5 as the height instead of 4 as the height of the parallelogram. Have students show how the triangle can be shifted to create a rectangle, then ask the student which value is the height of the rectangle.

- a. Fill in the blanks to find the area of the parallelogram.

$$\text{Area of parallelogram} = b \cdot h$$

$$\text{Area} = 6 \cdot 4$$

$$\text{Area} = 24 \text{ units}^2$$

- b. Three lengths are shown in the figure. With your partner, discuss the length that was not used to find the area of the parallelogram. Then, write an explanation for why this length was not used.
Answers will vary. The length on the slanted side is not perpendicular to the base therefore it is not the height, and the height must be used to find the area.

7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid

Component Overview: Students decompose a trapezoid into two triangles and use the triangles to derive the area formula for trapezoids. Encourage students to discuss their strategies with a partner.



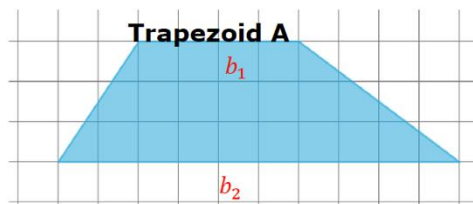
7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid

A trapezoid is another type of quadrilateral.



A **trapezoid** is a quadrilateral with at least one pair of parallel sides.

Trapezoid A is shown on a grid with each gridline representing one unit.



The two parallel sides of a trapezoid are both called bases. To distinguish between the two bases, one is named base one, b_1 , and the other base two, b_2 .

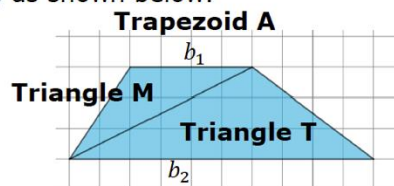
1. On Trapezoid A, label one of the bases b_1 and the other b_2 .
Shown on diagram.
2. Complete each statement.
 - a. The length of b_1 of Trapezoid A is 4 units.
 - b. The length of b_2 of Trapezoid A is 10 units.
 - c. The height of Trapezoid A is 3 units.



Clarify that students can label either base as b_1 and b_2 . Their labeling choices will impact their problem 2 answers.

7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid (continued)

3. Trapezoid A is decomposed into two triangles and labeled, as shown below.



- a. Complete the statements.
The base of Triangle M, b_1 , is 4 units, and its corresponding height is 3 units.
The base of Triangle T, b_2 , is 10 units, and its corresponding height is 3 units. When a trapezoid is decomposed into two triangles by drawing a diagonal, the height of both triangles will always be the same.
- b. Davita wrote a formula for the area of Trapezoid A.

Davita's Equation

Area Trapezoid A = Area Triangle M + Area Triangle T

$$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$$

Explain how Davita determined this formula.

Answers will vary. Davita realized that the area of the trapezoid must equal the area of the two triangles; so she added together the formulas for the areas of both of the triangles to get the area of the trapezoid.

- c. The formula Davita wrote can be used to find the area of any trapezoid. Complete the steps described in the table to rewrite the formula.

Area of Trapezoid A	
Davita's formula	$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$
Use the distributive property to rewrite with a common factor h	$A = h\left(\frac{1}{2}b_1 + \frac{1}{2}b_2\right)$
Use the distributive property to rewrite with a common factor $\frac{1}{2}$	$A = \frac{1}{2}h(b_1 + b_2)$



Prompt students to make connections to their decomposing in 7.4.2 Refresher.

- How is this the same as the Refresher?
- How is it different?
- What does it make you think about what the Area formula might be?



Students may struggle with understanding that the distributive property is applied. This is because the distributive property is often taught as distributing a value not factoring a value. It may be helpful to show the students $10 + 15 = 5(2 + 3)$ as the distributive property. Properties should be taught and used "both ways". The algebra in rewriting Davita's formula should not be the focus of this station. If you think it will be a stumbling block for your students then provide them with the answers. It is only here to show students how what they derived is equivalent to the formula that is often provided.

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus

Component Overview: Students derive the formula for the area of a rhombus by decomposing it into two triangles. Continue to encourage students to collaborate with partners, making sense of their discoveries as well as any connections to area.



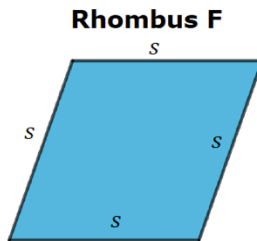
7.4.5 Activity: Deriving the Formula for the Area of a Rhombus

A rhombus is another type of quadrilateral.

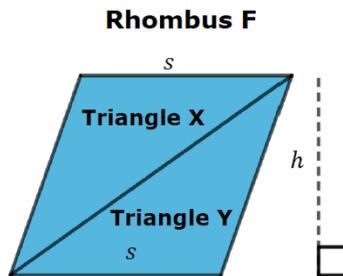


A **rhombus** is a quadrilateral containing four sides of equal length.

Rhombus F is shown with four congruent sides.



Rhombus F is decomposed into two triangles, Triangle X and Triangle Y, as shown.



Students may struggle with deriving the formula for the area of a rhombus using the two diagonals. The derivation of this formula extends student understanding both geometrically and algebraically. Be prepared with support for students as they complete this activity.



Prompt students to make connections to their decomposing in 7.4.2 Refresher.

- How is this the same as the Refresher?
- How is it different?

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus (continued)

1. Felicia wrote expressions to represent the areas of the two triangles.

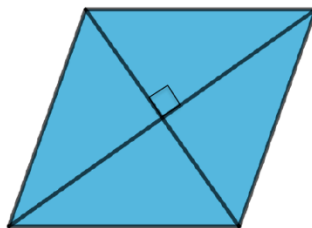
Area of Triangle X	Area of Triangle Y
$\frac{1}{2} \cdot s \cdot h$	$\frac{1}{2} \cdot s \cdot h$

Discuss with your partner whether you agree with the expressions Felicia wrote. Write a summary of your discussion.

Answers will vary. I agree with the expressions Felicia wrote because all of the sides are the same length on both triangles; which makes the triangles congruent and the formula to find area the same.

The diagonals of a rhombus can also be used to find its area.

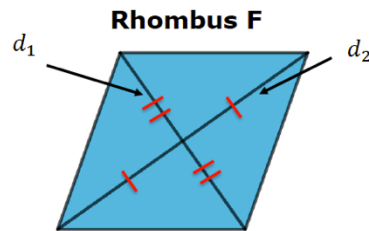
The two diagonals of a rhombus are perpendicular to each other and divide each other in half.



Is s the same for both triangles? How do you know? Is h the same for both triangles? How do you know? Could the formula for the rhombus be derived the same way the formula for parallelograms was derived in the 7.4.3 Activity?

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus (continued)

2. The two diagonals, d_1 and d_2 , of Rhombus F are drawn and labeled below.



Mark on the figure the two lengths that are equal to $\frac{1}{2}d_1$ and the two lengths that are equal to $\frac{1}{2}d_2$.

3. The first column of the table shows Rhombus F with parts highlighted. Use the lengths marked on Rhombus F in the previous question to fill in the blanks to find the area of the triangles.

Triangle	Area
Rhombus F 	$\text{Area} = \frac{1}{2}(b)(h)$ $\text{Area} = \frac{1}{2}(d_2)(\frac{1}{2}d_1)$ Rewrite the area by multiplying any fractions. $\text{Area} = \frac{1}{4}d_2d_1$
Rhombus F 	$\text{Area} = \frac{1}{2}(b)(h)$ $\text{Area} = \frac{1}{2}(d_2)(\frac{1}{2}d_1)$ Rewrite the area by multiplying any fractions. $\text{Area} = \frac{1}{4}d_2d_1$

4. Compare your work with your partner and resolve any differences, if necessary.

Answers will vary.



Which diagonal represents the base of the triangles in the table? Which diagonal represents the height of the triangles in the table?



Consider having students use different colored highlighters to trace over d_1 and d_2 in the figures and ask them to label d_1 and d_2 in the table.

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus (continued)

5. Work with your partner to express the area of Rhombus F as the sum of the area of the yellow triangle and the pink triangle found in the table.

$$\text{Area of Rhombus F} = \frac{1}{4}d_2d_1 + \frac{1}{4}d_2d_1 = \frac{1}{2}d_2d_1$$

6. Use words, values, or expressions from the bank shown to complete the statements. Items from the bank may be used more than once.

$\frac{1}{2}$	sum	$\frac{1}{4}(d_1)(d_2)$	diagonal
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Drawing a **diagonal** of a rhombus divides the rhombus into two identical triangles. The area of the rhombus is equal to the **sum** of the areas of these two triangles. If one of these triangles is equal to $\frac{1}{4}(d_1)(d_2)$, then two of these triangles equal

$2 \cdot \frac{1}{4}(d_1)(d_2)$. Therefore, the formula for the area of a rhombus using its diagonals can be written as

$$A = \frac{1}{2}(d_1)(d_2).$$



Encourage students to write out their work in problem 5. Prompt students to look for like terms when adding. Remind students to look back at problem 3 when completing problem 6. Have students pay particular attention to the area column, as it highlights the area of one of the triangles created by drawing in a diagonal.



Consider asking students who finish early to apply the formulas they developed to create problems involving the areas of parallelograms, trapezoids, or rhombi. Then, have students swap problems with a partner and solve their partner's problems.

7.4.6 Wrap-Up: Reflection

Component Overview: Students reflect on the most difficult or confusing point in the lesson. Use data from this formative assessment to determine which students need remediation before the next lesson.



7.4.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point for you in today's lesson on deriving the formula of parallelograms, trapezoids, and rhombi?



Answers will vary. The muddiest point for me in today's lesson is deriving the formula for rhombi.



Consider allowing students to use an audio recorder (such as Flip Grid) or speech to text software if they struggle expressing themselves in writing. Students could also create a code (such as 0 = do not understand, 1 = sort of understand, 2 = completely understand) and use it to label the problems in the lesson.